

Pranay Oza

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SKILLS

Programming & Software: C, C++, Assembly, SystemVerilog, Rust, Python, Java, MATLAB, Git, Unix, CMake

Embedded Firmware: STM32, FreeRTOS, FDCAN/CAN, SPI/isoSPI, I²C, UART, USB, OCPP, J-Link

Hardware & Systems Design: Altium, KiCad, Power Electronics, Circuit Analysis, Simulink, LTspice

EDUCATION

University of British Columbia

Computer Engineering

Expected Graduation – Dec 2028

Vancouver, BC

EXPERIENCE

High Voltage Firmware Integration

Tesla

Incoming May 2027

Palo Alto, CA

Battery Management System Firmware Lead

UBC Formula Electric SAE

Sept 2024 – Present

Vancouver, BC

- Leading firmware development for the car's fully custom **600V BMS**, owning the full stack from high-level **battery algorithms** down to low-level drivers monitoring 500+ 18650 lithium-ion cells
- Developing a driver for the **ADBMS2950** to monitor tractive-system voltage and pack current across a **shunt resistor**, providing the isolated high-voltage measurements required for accurate power and energy tracking
- Modeling and validating a **second-order RC** equivalent-circuit battery model in **MATLAB**, parameterized from **HPPC**, **OCV–SOC**, and capacity characterization tests ran on a battery cycler, improving fidelity of state estimation
- Researching battery state-estimation algorithms spanning **state-of-charge**, **state-of-health**, **state-of-energy**, and **state-of-power**, giving greater visibility into pack condition and to inform energy strategy
- Building **Rust** tooling that auto-generates **multiplexed FDCAN signals** from a single JSON definition and integrates with the telemetry backend, packing more data into FDCAN messages to raise logging density without adding bus load
- Relocating **ISR** and the **interrupt vector table** into **Tightly-Coupled Memory**, minimizing instruction-fetch latency to achieve deterministic, lowest-possible interrupt response times
- Validated safety-critical fault handling by injecting faults into the BMS and confirming each tripped the pack's **shutdown loop**, covering insulation-monitoring, brake-system, and inter-segment faults to ensure the pack fails safe
- Developed **CAN-based charging firmware** for a 6.6 kW Elcon charger with a full CC/CV profile and termination logic, and integrated **J1772 EVSE** charging by decoding the control-pilot duty cycle to safely limit the current request
- Implemented a **FDCAN-based bootloader** with **CRC32** verified-boot to flash all boards over FDCAN without a debugger, using **flash-memory partitioning** to keep every node on a version-matched build
- Built an **automated calibration script** and lab-bench setup for the pack current sensor, delivering the accurate power measurement the vehicle-dynamics team relied on to set the car's power limit
- Owned bring-up and on-hardware validation of the firmware on the team's custom BMS PCBs, verifying reliability through hardware testing and **software-in-the-loop (SIL)** simulation before vehicle integration

PROJECTS

ADBMS6830B Driver Development – *UBC Formula Electric SAE*

Jan 2026 – Sept 2026

- Owned the architecture, implementation, and integration of a driver for daisy-chained **ADBMS6830B** monitor ICs (SPI/isoSPI) across a multiple **56 cell segments** with **PEC15/PEC10 CRC** frame validation
- Implemented **cell-voltage** and **thermistor** sensing, **open-wire** detection, and per-cell **balancing** with **reversible isoSPI** and dropped-command detection for daisy-chain fault tolerance, organized into reusable modules behind a **hardware-abstraction layer** that broadcasts per-segment fault flags over FDCAN
- Designed a **FreeRTOS** concurrency model with independently clocked config, voltage, open-wire, and thermistor tasks sharing one SPI bus through **synchronization primitives**, eliminating bus contention and data races
- Debugged **15+ boards** and daisy chain for signal-integrity, hardware shorts, and DMA/SPI/task-scheduling bugs using **SEGGER SystemView** for RTOS task visibility, then ran full-battery integration testing

Sensorless Field-Oriented Control ESC

Jun 2025 – Present

- Designing a **4-layer PCB** integrating an STM32G4 MCU, gate driver, buck converter, EEPROM, ESD protection circuitry, and reverse polarity detection for 30A, 12V BLDC motor control
- Developing motor-control firmware (PlatformIO, STM32CubeMX) using ADC sampling, PWM timers, I²C, and UART for current sensing, **virtual neutral-voltage** measurement, and **back-EMF** zero-cross detection
- Building a Simulink inverter model to simulate 3-phase dynamics and validate **sensorless FOC** algorithms, including SVPWM and dead-time compensation